taken from a large consignment and 65 were found to be bad.

6. A sample of 100 students are drawn from a school. The mean weight and variance of the sample are 67.45 kg and 9 kg. respectively. Find (a) 95% and (b) 99% confidence intervals for estimating the mean weight of the students.

7. The mean I.Q of a sample of 1600 children was 99. Is it likely that this was a random sample from a population with mean I.Q 100 and standard deviation 15 ? (Test at 5% level of significance).

Summary

- **Sampling:** It is the procedure or process of selecting a sample from a population.
- **Population:** The group of individuals considered under study is called as population.
- **Sample :**Aselection of a group of individuals from a population.
- **Sample size :**The number of individuals included in a sample.
- **Simple Random Sampling :**The samples are selected in such a way that each and every unit in the population has an equal and independent chance of being selected as a sample.
- **Stratified Random Sampling:** When the population is heterogeneous, the population is divided into homogeneous number of sub-groups or strata. A sample is drawn from each stratum at random.
- **Systematic Sampling:** Select the first sample at random, the rest being automatically selected according to some predetermined pattern.
- **Sampling Distribution:** Sampling distribution of a statistic is the frequency distribution which is formed with various values of a statistic computed from different samples of the same size drawn from the same population.
- **Standard Error:** The standard deviation of the sampling distribution of a statistic is known as its Standard Error.
- **Statistical Inference:** To draw inference about a population of any statistical investigation from the analysis of samples drawn from that population.
- **Estimation :**The method of obtaining the most likely value of the population parameter using statistic is called estimation.
- **Point Estimation:** When a single value is used as an estimate, it is called as point estimation.
- **Interval Estimation:** An interval within which the parameter would be expected to lie is called interval estimation.
- **Test of Statistical Hypothesis:** Statistical technique to arrive at a decision in certain situations where there is an element of uncertainty on the basis of sample
- **Null Hypothesis:** The hypothesis which is tested for possible rejection under the assumption that it is true", denoted by H_0.
- **Alternative Hypothesis:**The hypothesis which is complementary to the null hypothesis is called as the alternative hypothesis, denoted by H_1.
- **Type I error:**The error of rejecting H_0 when it is true.
- **Type II error:**The error of accepting H_0 when it is false.
- **Test of significance for single mean:**

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$$

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